

Data Analyst Project

TikTok Performance Analysis Across Seven Indonesian Media Accounts

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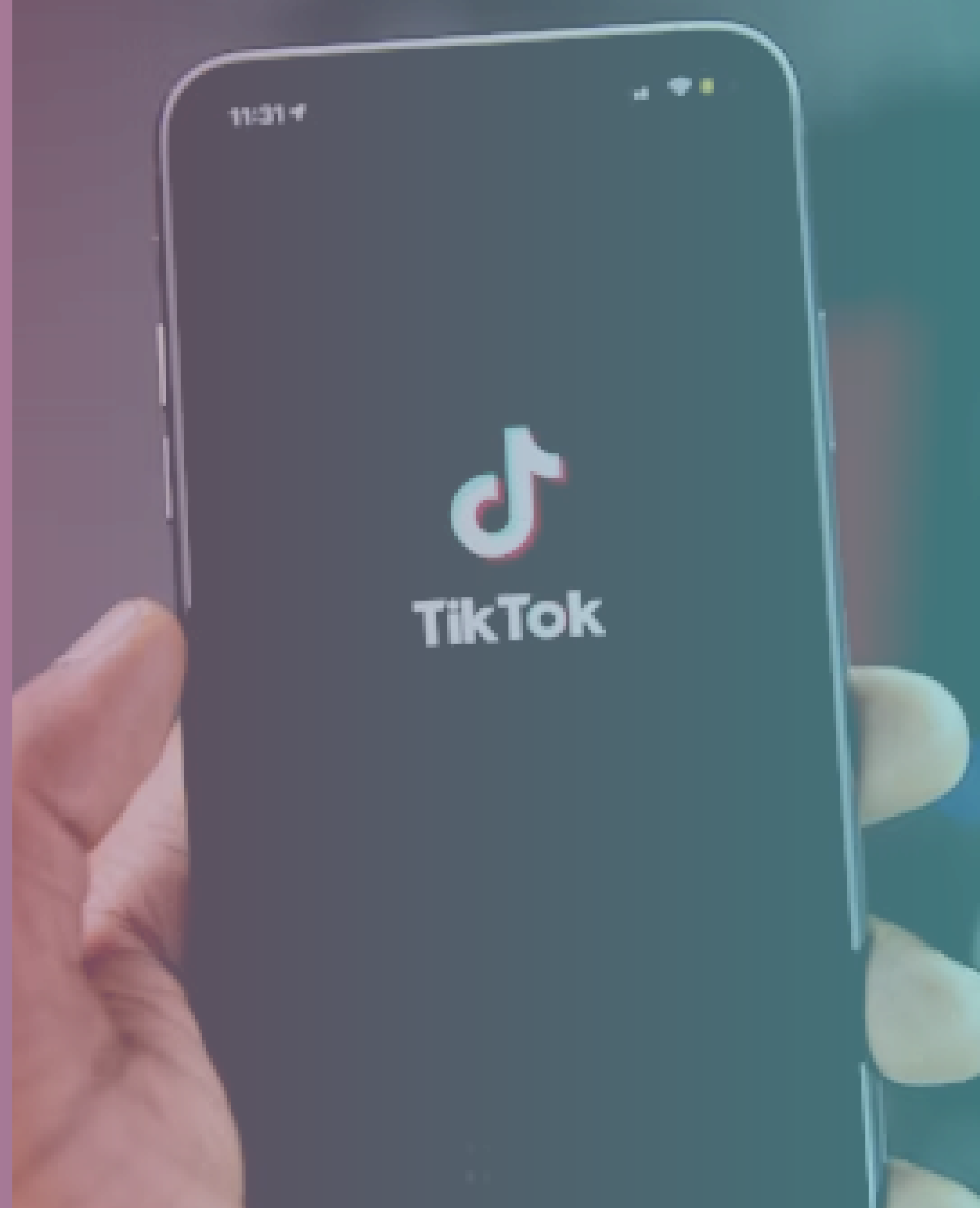
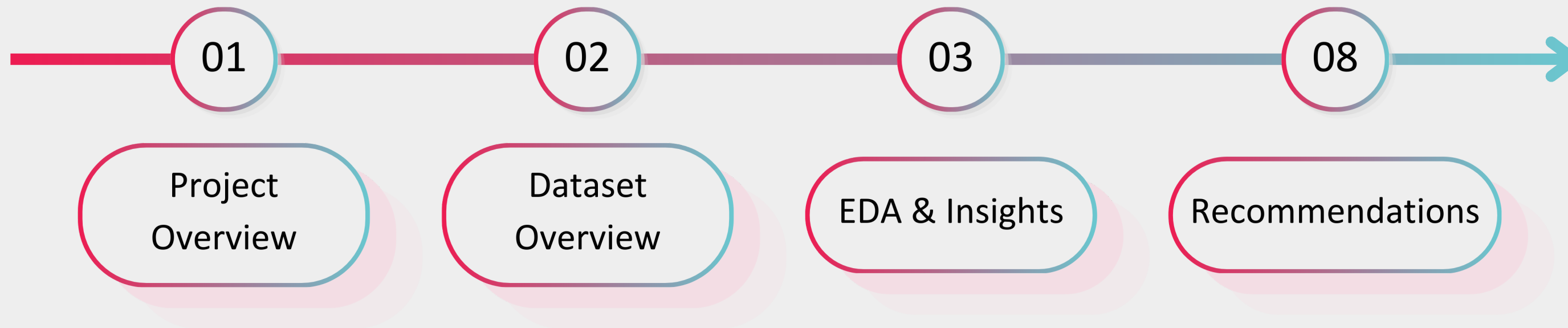


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Understanding the business problem, objectives, and analytical framework

Business Understanding

TikTok has become a key platform for news, where Indonesian media compete in a fast-paced content environment. However, performance remains inconsistent due to short-term virality and the lack of data-driven strategies, making it difficult to optimize posting strategies to consistently maximize reach and engagement.

Objectives

- 01 Analyzed TikTok data to identify impressions, engagement trends, content performance, and posting behavior using Python.
- 02 Build a simple dashboard in Google Data Studio for data visualization.
- 03 Provide data-driven recommendations to improve content strategy.

Framework

Data Collection

Obtain the TikTok dataset from SocialInsider and explored 4,275 rows × 23 columns.

Data Preparation

Process, transform, clean, and standardize the raw dataset into an analysis-ready format.

Exploratory Data Analysis

Identify impressions and engagement trends, content performance, and posting behavior.

Dashboard Development

Build an interactive dashboard in Google Data Studio.



Insights Generation

Overview dataset structure, quality, and variables

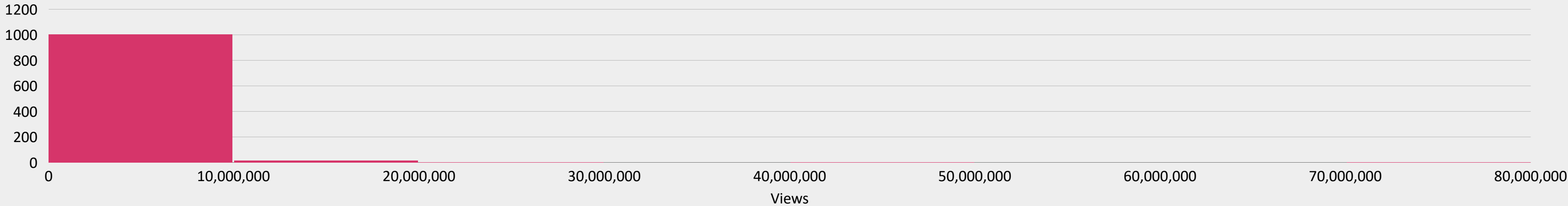
Total Rows	Total Column
4,275	23
Total Missing Values	Total Duplicate Rows
34,200	0
Source	Period
SocialInsider	26 May–24 June 2026
Additional Notes	
• The dataset relies on third-party data from SocialInsider, so its accuracy and completeness can't be fully guaranteed. • Limited access to advanced metrics (such as audience demographic and follower growth) may affect the depth of analysis.	

Data Flow
Data Issues
• There are 4 columns with 100% missing values. • There are 2 columns with partial and minor missing values. • All columns with missing values are not relevant to the analysis. • No duplicate rows found.
Data Cleaning
• Removed irrelevant columns that contained missing values. • Retained only relevant columns for performance analysis. • The dataset is clean and ready for analysis (4,275 rows × 15 columns).
Data Analysis & Visualization
• The cleaned dataset will be exported as a CSV file from Google Sheets. • The file will be used to build dashboard in Google Data Studio, then EDA will be conducted using this dashboard.

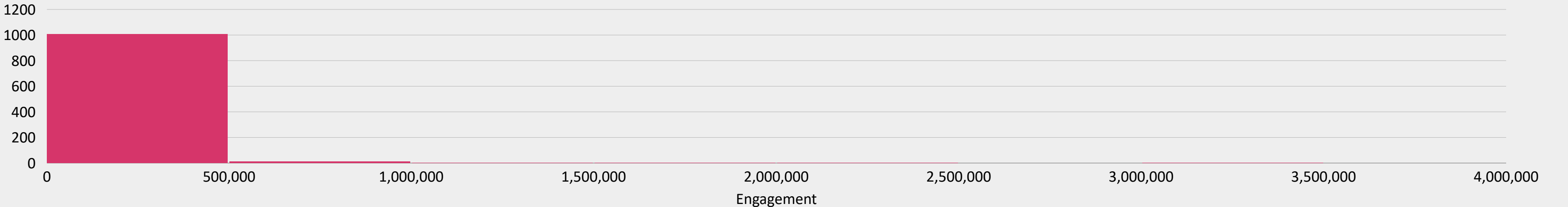
Examining the distribution of key performance metrics before interpretation and further analysis



Views Distribution



Engagement Distribution

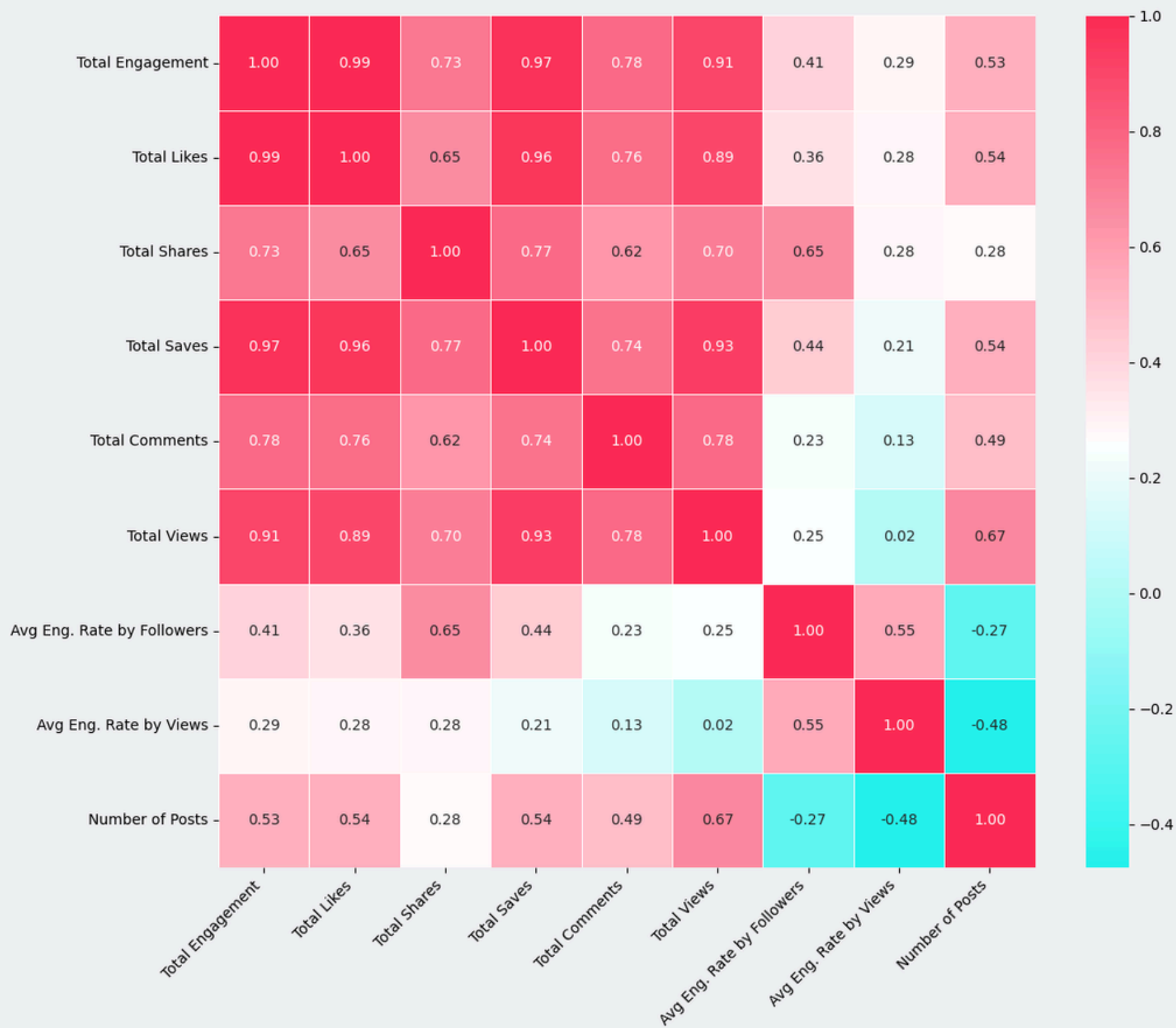


Key Insights

- ✓ The distributions of both views and engagement are **heavily right-skewed**, indicating that performance is concentrated in a small number of high-performing posts rather than being evenly distributed.
- ✓ Given this skewness, the **median provides a more reliable measure**, as it minimizes distortion from viral outliers and better represents the typical content performance.

Daily metrics correlation heatmap to understand relationships between metrics

Daily Metrics Correlation Heatmap



Daily Metrics Correlation Heatmap



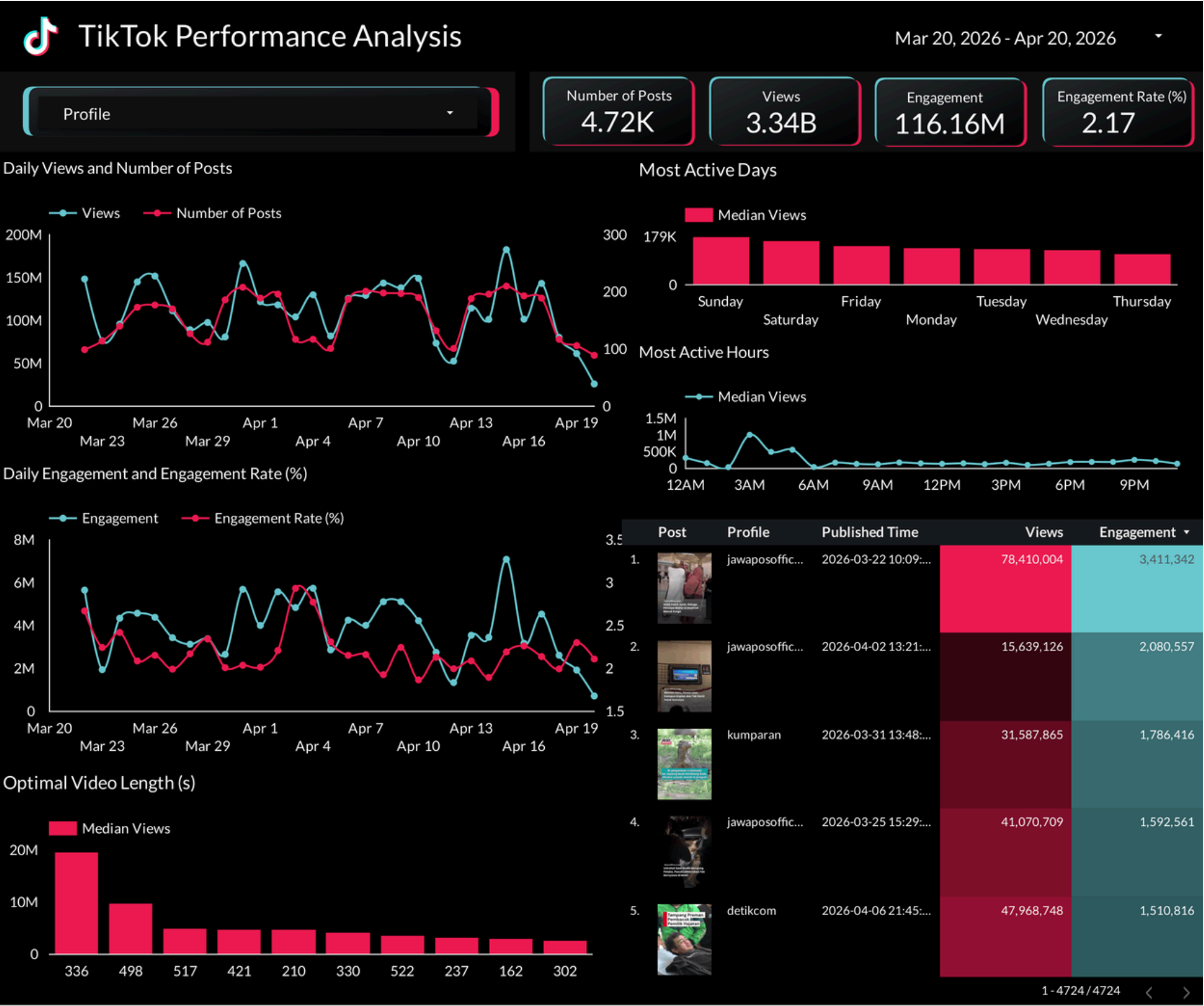
Total Views and Total Engagement showed a very strong positive correlation (0.91), suggesting that higher visibility content consistently translated into higher engagement.



Among engagement metrics, Total Saves (0.93) and Total Likes (0.89) were the strongest correlated with Views, indicating they were the most reliable indicators of content success.



Number of Posts had a moderate correlation with Total Views (0.67) and Total Engagement (0.53), implying that posting frequency contributed to reach and engagement, but was not the primary performance driver.



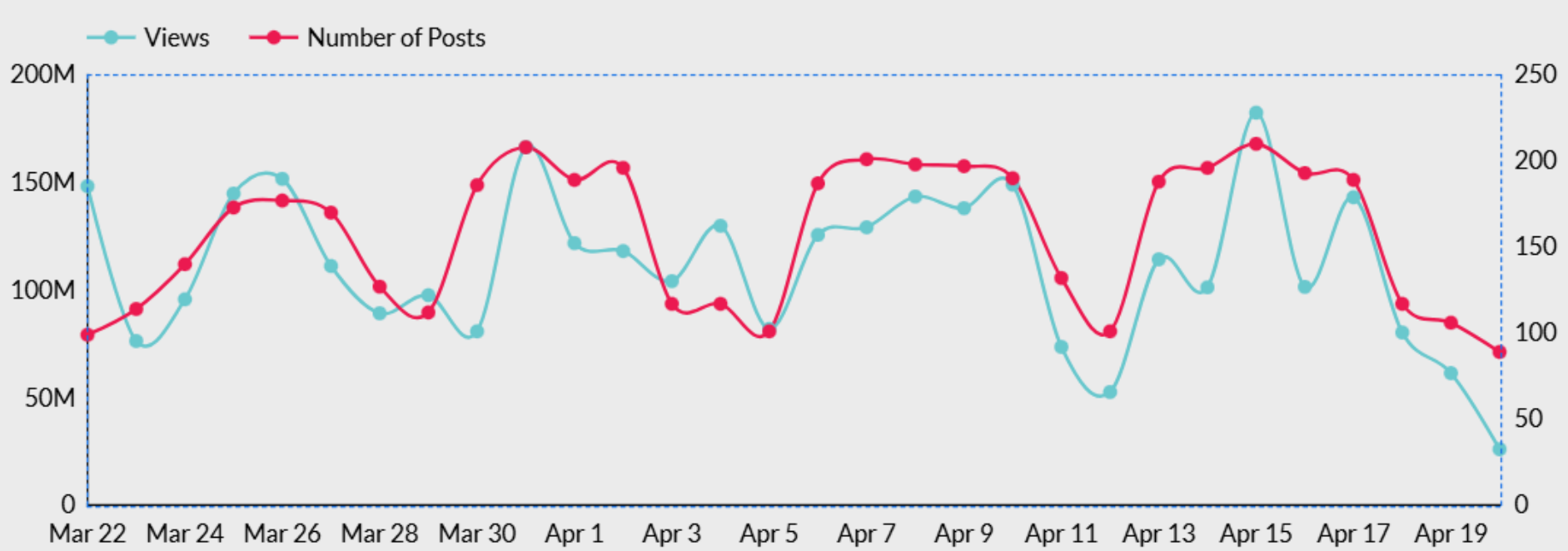
Link Dashboard



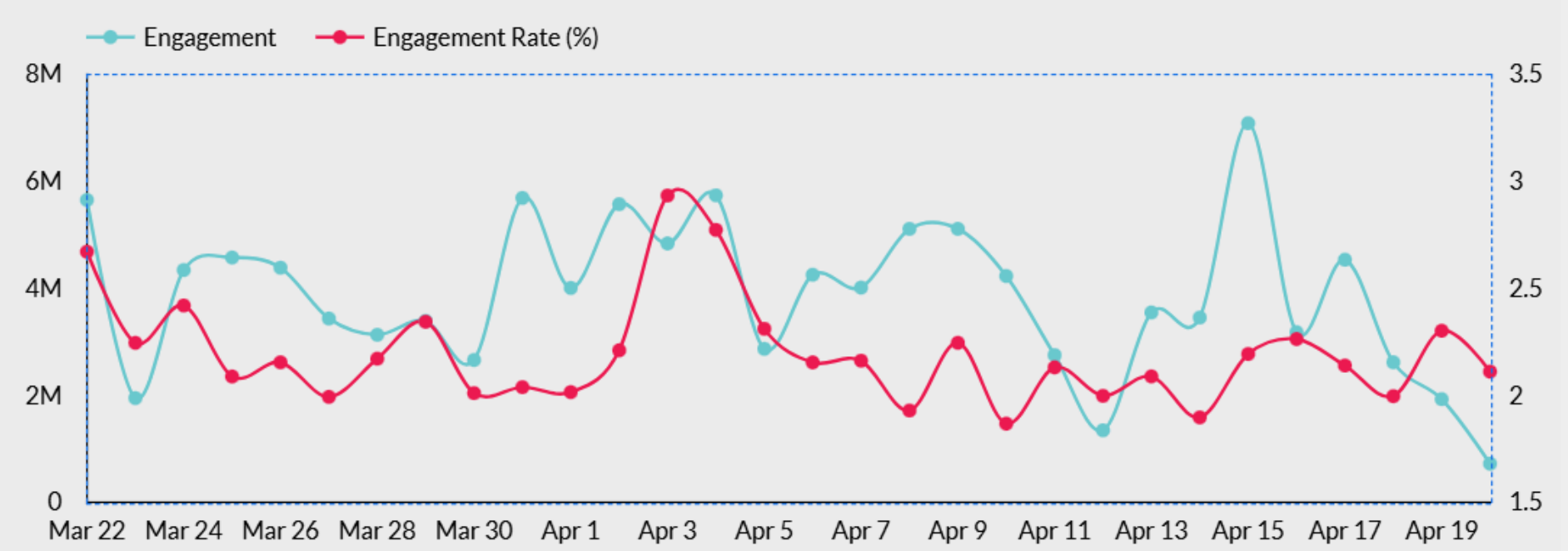
- Over a 30-day period, **overall content output was high** at 4.72K posts, generating 3.34B views and 116.16M engagements, but with a relatively low average engagement rate of 2.17%. This indicated limited user actions beyond viewing.
- Kumparan led in both volume and performance**, with 1.00K posts driving 902.14M views and 30.21M engagements, indicating that higher views correlated with higher engagement.
- Pikiran Rakyat recorded the lowest performance**, with 59.16M views and 1.71M engagements, suggesting comparatively weaker content effectiveness across media.

Examining the daily trend of views, engagement, and number of posts

Daily Views and Number of Posts



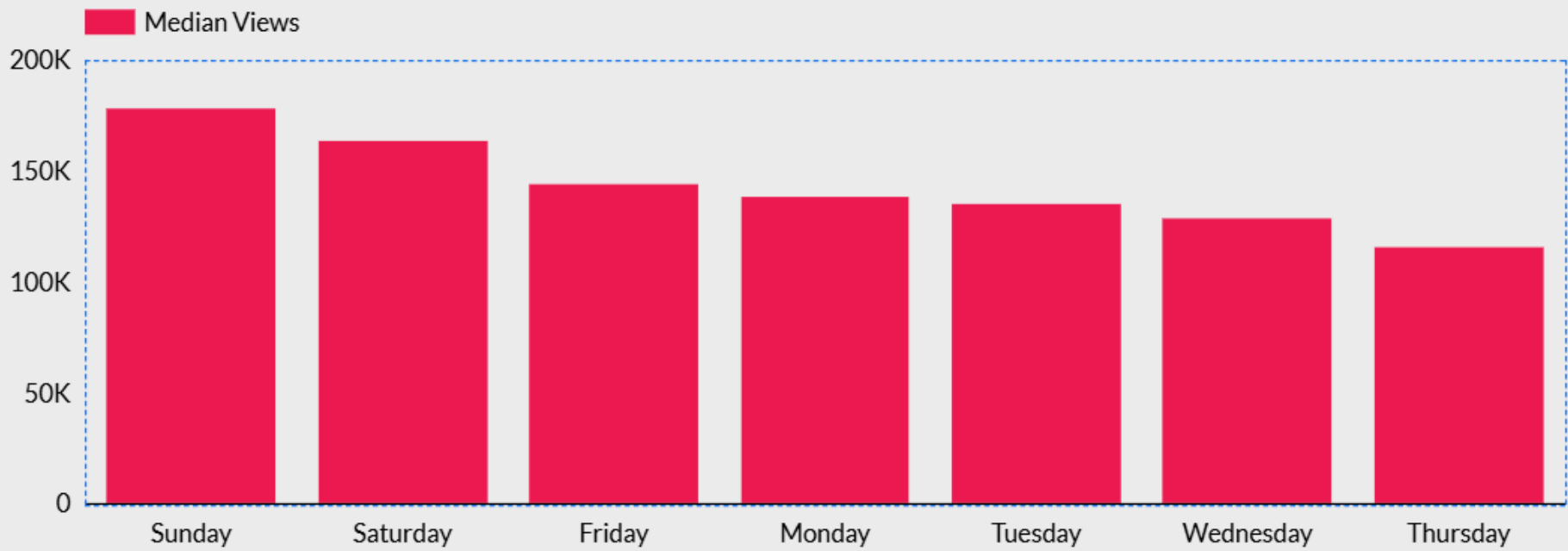
Daily Engagement



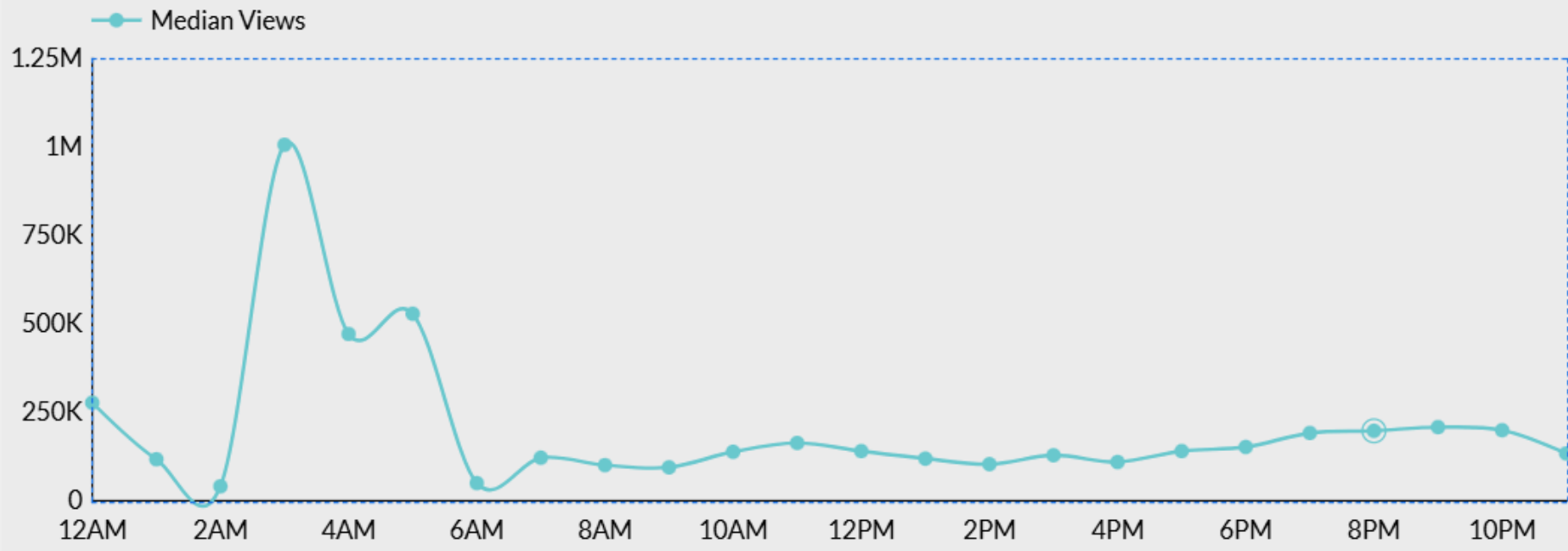
Insights

- Number of Posts and Views did not show a consistent correlation, indicating that **posting frequency alone was not a reliable driver of reach**.
- Engagement and Engagement Rate were not consistently aligned, suggesting that **higher reach did not always translate into stronger audience interaction**.
- Performance over time was uneven, with April 15 as the peak for Views, Posts, and Engagement, indicating a likely **event-driven spike** in performance.
- In contrast, April 20 had the lowest Engagement, while April 14 recorded the lowest Engagement Rate, showing fluctuating audience responsiveness across days.

Prime Days



Prime Time

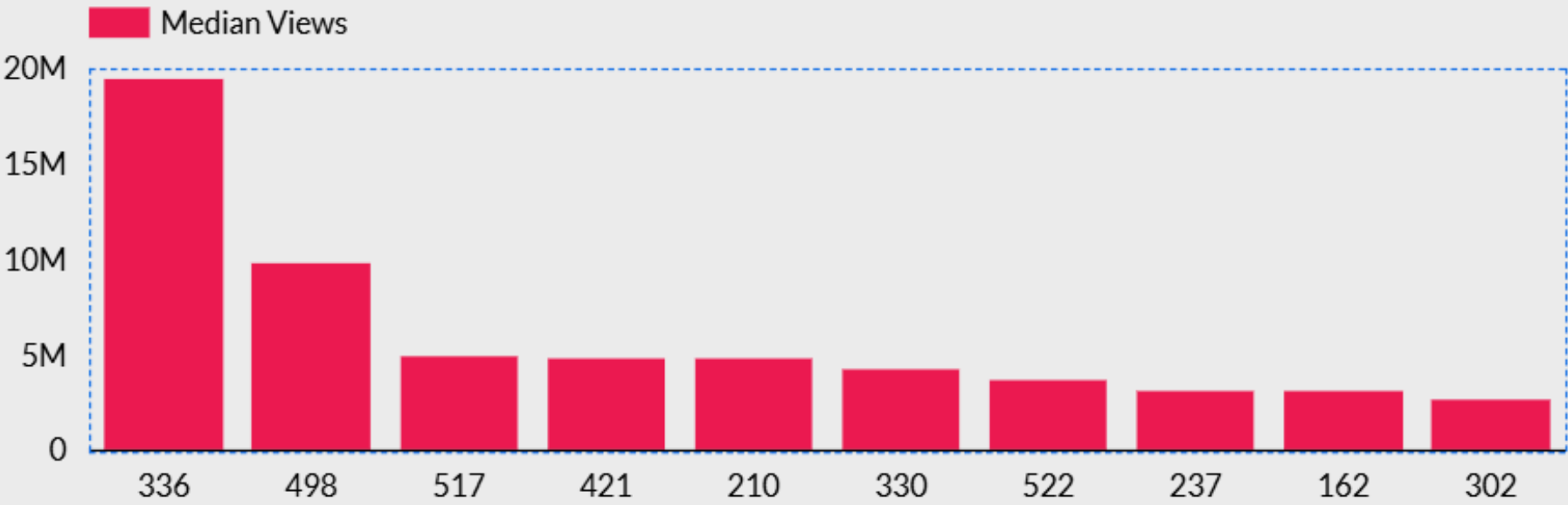


Insights

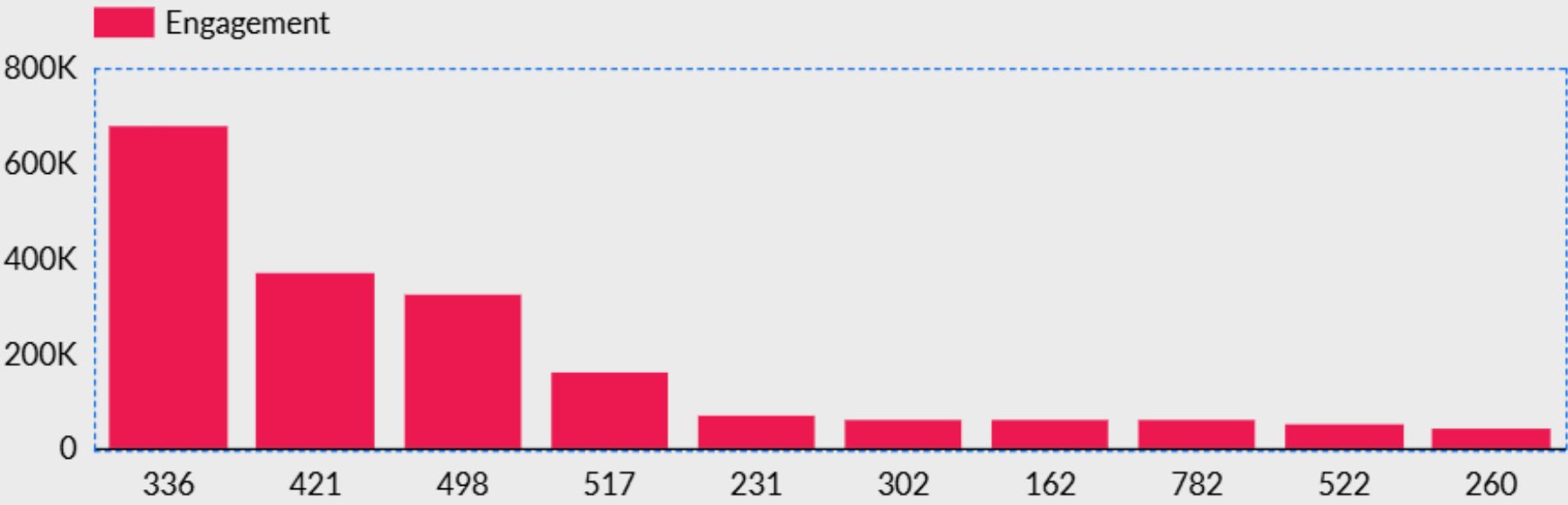
- **Weekend postings (Saturday–Sunday) showed higher median views**, indicating better performance compared to weekday uploads.
- **Posts published between 3–5 AM showed the highest median views**, indicating this time window consistently delivered stronger reach performance.

Examining how video length imapcts median views and engagement

Optimal Video Length (s) by Median Views



Optimal Video Length (s) by Median Engagement



Insights

- Content around 336 seconds showed the highest median views and engagement, indicating that **audiences responded well to extended, well-developed storytelling**.
- Very long content (>336 seconds) and shorter–mid content (<336 seconds) showed noticeably lower performance, suggesting **shorter videos may have underdelivered value, while longer ones risked audience fatigue**.

Top 5 Content by Views



01



02



03



04



05

Key Insights

Top-performing content was largely **momentum-driven**, as most topics were tied to recent or ongoing events.

The best-performing content was led by International topics, indicating **audiences were highly attracted to globally relevant issues** involving well-known figures, especially when combined with controversy or unusual statements.

Disaster and Accident dominated the top 5 (2 out of 5), showing that **content with urgency, shock value, or dramatic visuals** was more likely drive high views.

Top 5 Content by Engagement



01



02



03



04



05

Key Insights

Top engagement content was also **momentum-driven, dominated by recent issues like geopolitics and economic shifts**. However, it skewed more toward topics with personal impact or interpretive value, which were more likely to trigger active responses.

The best-performing content was also led by International topics, while Business and Economic topics dominated the top 5 (2 out of 5), indicating **audiences were more likely to engage with content about economic shifts (gold and fuel prices) due to their direct relevance to everyday life**.

Recommendations	Key Finding	Recommended Action	Expected Impact
 Prioritize Momentum-Driven Content	Top-performing content was largely momentum-driven, as most topics were tied to recent or ongoing events.	Focus on momentum-driven topics (e.g., international politics, disasters, and economic) as since they consistently drive both high views and engagement.	Higher views and engagement.
 Prioritize Posting on Prime Day and Time	Videos published at weekends and 3–5 AM generated higher median views.	Optimize posting timing, especially weekends and early morning (3–5 AM), which showed stronger median views performance.	Better audience exposure and interaction also higher views potential.
 Prioritize Medium to Long Duration Videos	Video with medium to long duration (around 336 s) generated higher median views and engagement.	Standardize video duration around 336 seconds, decrease production of short and very long video.	Higher views potential.
 Prioritize Content Quality over Quantity	Number of posts daily had moderate correlation with views and engagement.	Shift strategy from posting volume to content quality and relevance, since frequency alone has weaker impact compared to topic strength and content effectiveness.	More effecient resources allocation and sustainable views and engagement growth.

LET'S CONNECT



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